

Kyle Main

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Senior Machine Learning Engineer & Security Data Scientist — Anomaly Detection, Clustering & Detection-as-Code

Security data scientist and ML engineer with 12 years of experience building unsupervised and time-series anomaly detection systems that run in production against massive-scale security telemetry — clustering network devices and entities by behavior, detecting anomalous authentication and process-chain patterns, and running exploratory analysis at scale on PySpark/Dataproc/BigQuery. Built and operate a multi-SIEM detection-as-code CI/CD pipeline (GitLab) with automated testing, staged/safe rollout, and rule-quality metrics tracking before production — the same engineering rigor this role asks for, already applied at scale (2,300+ rules across the MITRE ATT&CK matrix). M.S. Physics underpins the quantitative foundation.

CORE SKILLS

Machine Learning & Data Science: Clustering/unsupervised ML for device and entity behavior; time-series anomaly detection (authentication volume/velocity, process-parent/child chains); exploratory data analysis at scale; Python, PyTorch, scikit-learn, pandas, NumPy, SciPy

Big Data & Distributed Compute: PySpark, SparkSQL, GCP Dataproc, BigQuery, Apache Beam/Dataflow; working familiarity with Kafka and Flink streaming; cloud: GCP, AWS, Azure

MLOps / Detection-as-Code Pipelines: GitLab CI/CD pipeline for detection-as-code with automated unit/integration tests, staged/safe rollout, and formally tracked rule-quality metrics (coverage, precision/false-positive rate) before production deployment; multithreaded orchestration across customer environments

Containers & Orchestration: Docker (built and managed custom images; containerized reproducible detection-testing environments); hands-on experience operating within Kubernetes-orchestrated platforms

Cybersecurity Domain Depth: MITRE ATT&CK-mapped detection content (2,300+ rules); multi-SIEM API orchestration (Splunk, Microsoft Sentinel/Defender, Google SecOps, CrowdStrike, SentinelOne, Sumo Logic, XSIAM, Devo, ArcSight); threat intel integration into detection logic

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Design and manage statistical and behavioral detection analytics (Splunk saved searches) directly supporting Treasury's Security Operations Center incident response and case work (Treasury SOC / TSSOC, current project).
- Architected an unsupervised UEBA (User and Entity Behavior Analytics) detection layer from scratch for DOE/NNSA — Elasticsearch transforms baselining entity behavior across CrowdStrike, Suricata, and Zeek telemetry, plus anomaly-surfacing content, data-quality monitoring, and custom dashboards (DOE/NNSA Security Data Integration, completed) — turning a vague threat-visibility mandate into a concrete production system.
- Supported data ingestion and data-quality efforts within an Elasticsearch/Splunk environment for DOE's Continuous Diagnostics and Mitigation (CDM) program (completed).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and detection engineering team building statistical, behavioral, time-series, and ML-based (clustering) detection models against massive-scale customer telemetry, with a direct mandate to reduce alert fatigue and improve analyst trust.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Built unsupervised clustering models to group devices on the network by behavioral feature for anomaly surfacing, and time-series anomaly detection covering authentication behavior (anomalous volume/attempt patterns by country over time) and process-chain behavior (parent/child process combinations), as core production detection content.
- Ran exploratory data analysis at scale on GCP Dataproc (Zeppelin notebooks, a home-grown reusable analysis toolkit, PySpark/SparkSQL) to mine 220+ ingested log sources for signal.
- Created and managed 2,300+ individual detection rules covering most of the MITRE ATT&CK matrix as a very early hire, building the rules engine and detection content for the startup from scratch — plus the data engineering (50+ Logstash filters, Common Information Model) underneath it.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Developed custom statistical models and algorithms for DNS-based detection and mitigation of malware infections on the network, and analyzed large-scale security log data to surface anomalous behavior.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox

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August 3, 2026

Trust Intelligence Platform Hiring Team
Salesforce

Moving detection beyond signatures into probabilistic modeling and unsupervised learning is work I've already been doing for over a decade in production security environments — clustering entities by behavior, surfacing anomalies in time-series data, and building the pipelines that let that content actually ship. I'd like to bring that background to the Lead Machine Learning Engineer role on Salesforce's Trust Intelligence Platform team.

At Trend Micro/Cysiv and Forescout, I built unsupervised clustering models to group devices on the network by behavioral feature, and time-series anomaly detection covering authentication patterns and process-chain behavior — production detection content, not research prototypes, running against massive-scale customer telemetry and covering most of the MITRE ATT&CK matrix across 2,300+ individual rules. At DOE/NNSA, I took a vague mandate — give the SOC visibility into unknown threats — and turned it into a concrete system: an unsupervised UEBA detection layer built on Elasticsearch transforms I designed from scratch, exactly the kind of translating-ambiguity-into-a-data-driven-solution this role calls for. All of this exploratory and model-development work ran at scale on GCP Dataproc, PySpark, and SparkSQL.

I've also already built the operational discipline this role's MLOps mandate asks for: a multi-SIEM detection-as-code CI/CD pipeline in GitLab, orchestrating rule and content deployment across Splunk, Microsoft Sentinel, Microsoft Defender, Google SecOps, CrowdStrike, and more via their native APIs, with automated testing, staged/safe rollout, and formally tracked quality metrics before anything reaches production — the same rigor that keeps a SOC trusting the models it's given. I've built this as a force multiplier for other engineers, not just for myself, developing reusable per-platform adapters and tooling the whole detection team relies on. My M.S. in Physics underpins the quantitative side of all of it.

I'd welcome the chance to bring that same combination of production ML depth and engineering rigor to Salesforce's security data science team.

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